1. In *Drosophila,* the recessive allele *mb* of one gene causes missing bristles, the recessive allele *e* of a second gene causes ebony body color, and the recessive allele *k* of a third gene causes kidney-shaped eyes. (Dominant wild-type alleles of all three genes are indicated with a + superscript.) The three different P generation crosses in the table that follows were conducted, and then the resultant F 1 females from each cross were testcrossed to males that were homozygous for the recessive alleles of both genes in question. The phenotypes of the testcross offspring are tabulated as follows. Determine the best genetic map explaining all the data.

Parental cross	Testcross offspring of F_1 females	
$mb^+\ mb^+, e^+\ e^+ \times$ *mb mb, e e*	normal bristles, normal body	117
	normal bristles, ebony body	11
	missing bristles, normal body	15
	missing bristles, ebony body	107
$k^+\ k^+, e\ e \times k\ k, e^+\ e^+$	normal eyes, normal body	11
	normal eyes, ebony body	150
	kidney eyes, normal body	144
	kidney eyes, ebony body	7
$mb^+\ mb^+, k^+\ k^+ \times$ *mb mb, k k*	normal bristles, normal eyes	203
	normal bristles, kidney eyes	11
	missing bristles, normal eyes	15
	missing bristles, kidney eyes	193

① $mb^+mbe^+e \times mbmbee$

map distance : $\frac{11+15}{117+11+15+107} \times 100 = 10.4\ mu$

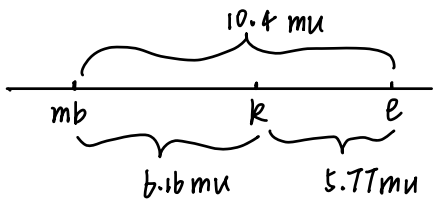
② $k^+ke^+e \times kkee$

map distance : $\frac{11+7}{11+150+144+7} \times 100 \approx 5.77\ mu$

③ $mb^+mbk^+k \times mbmbkk$

map distance : $\frac{11+15}{203+11+15+193} \times 100 \approx 6.16\ mu$

∴



2. A three-point testcross was made in corn. The results and a recombination analysis are shown in the display below, which is typical of three-point testcrosses (*p* = purple leaves, + = green; *v* = virus-resistant seedlings, + = sensitive; *b* = brown midriff to seed, + = plain). Study the display, and answer parts *a* through *c*.

P $+/+ \cdot +/+ \cdot +/+ \times p/p \cdot v/v \cdot b/b$

Gametes $+ \cdot + \cdot +$ $p \cdot v \cdot b$

a. Determine which genes are linked.
b. Draw a map that shows distances in map units.
c. Calculate interference, if appropriate.

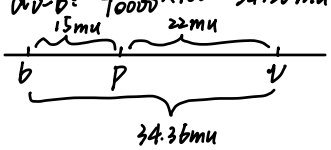
Class	Progeny phenotypes	F_1 gametes	Numbers	*Recombinant for*		
				p–b	*p–v*	*v–b*
1	gre sen pla	$+ \cdot + \cdot +$	3,210			
2	pur res bro	$p \cdot v \cdot b$	3,222			
3	gre res pla	$+ \cdot v \cdot +$	1,024		R	R
4	pur sen bro	$p \cdot + \cdot b$	1,044		R	R
5	pur res pla	$p \cdot v \cdot +$	690	R		R
6	gre sen bro	$+ \cdot + \cdot b$	678	R		R
7	gre res bro	$+ \cdot v \cdot b$	72	R	R	
8	pur sen pla	$p \cdot + \cdot +$	60	R	R	
		Total	10,000	1,500	2,200	3,436

a. p. v. b 三个基因都是 linked

b. $d_{p-b}: \frac{1500}{10000} \times 100 = 15\,mu$

$d_{p-v}: \frac{2200}{10000} \times 100 = 22\,mu$

$d_{v-b}: \frac{3436}{10000} \times 100 = 34.36\,mu$



c. $P_{double} = 0.15 \times 0.22$

$E = 0.15 \times 0.22 \times 10000 = 330$

$C = \frac{D}{E} = \frac{72+60}{330} = 0.4$

$I = 1 - C = 0.6$

2. In *Drosophila,* three autosomal genes have the following map:



a. Provide the data, in terms of the expected number of flies in the following phenotypic classes, when $a^+ b^+ c^+$ / *a b c* females are crossed to *a b c* / *a b c* males. Assume 1000 flies were counted and that there is no interference in this region.

a^+	b^+	c^+	360
a	b	c	360
a^+	b	c	90
a	b^+	c^+	90
a^+	b^+	c	40
a	b	c^+	40
a^+	b	c^+	10
a	b^+	c	10

b. If the cross were reversed, such that $a^+ b^+ c^+$ / *a b c* males are crossed to *a b c* / *a b c* females, how many flies would you expect in the same phenotypic classes?

a. $P_{a-b} = 0.2$, $P_{b-c} = 0.1$, $P_{double} = 0.2 \times 0.1 = 0.02$

$E_{double} = 0.02 \times 1000 = 20$

$(0.2 - 0.02) \times 1000 = 180$

$(0.1 - 0.02) \times 1000 = 80$

$1000 - (180 + 80 + 20) = 720$

(Expected number 标注在上图).

b. $a^+ b^+ c^+$ 500

$a\ b\ c$ 500